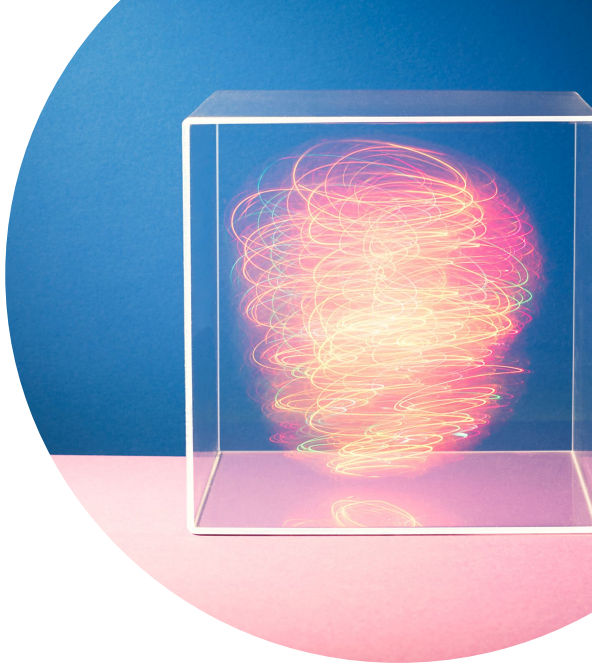
devoteam
G Cloud

# GCP Foundation
## Kick-off



OLENGEE

11/25/2024

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GCP Landing Zone

# Context

# Context - **Project purpose**

OLENGEE

**GCP Migration**
Migration to Google Cloud Platform (GCP) with the implementation of a standardized Cloud Foundation environment. The goal is to lay the foundation for a scalable and high-performing infrastructure.

# Objectives - **Deliverables**

**Kickoff**
**& weekly decks**

Initialization & Weekly project management:

- *Activities done*
- *Activities upcoming*
- *Key milestones*
- *Alerts*
- *Deliverables status*
- *Global project status*

**Workshops**
**& Minutes of Meeting**

- Governance & IAM
- Networking
- Security
- Billing
- Observability
- Automation
- VM provisioning

**Low Level Design**

- Technical documentation of the GCP environment which gather all the main validated decisions (Written in english)

**Build of the GCP Foundation**

- Deployment of GCP resources following workshops decisions
- Deployment automation with Terraform

**Final Restitution**

- Platform Review & Feedback
- Project Management status
- Next steps

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General Kick-off

# Governance

- Teams
- Working method

# Governance - **Project team**

## OLENGEE

### Sponsor

- F — François Roulin, Co-founder & CIO/CPO

### Technical project team

- D — Daniel Orozco, Project Manager
- M — Martin Chuburu, Technical Team Responsible
- F — Francisco Gutierrez, Google Platform Responsible

## devoteam G Cloud

### Business relationship

- A — Alexis Secher, Account Manager

### Technical project team

- H — Houssem Hamzaoui, GCP Architect
- H — Hae Jin Lim, Cloud engineer
- A — Aymeric Martin, Project Manager

# Governance - **Working methods**

### Share

- A Shared Drive folder will contain all documents in a simple two structure: Project / Technical.
- All our documents will be shared for collaborative work.

### Validate

- Documents issued will be proofread and validated within 10 days.

### Chat

- Regular exchanges will be encouraged wherever possible.
- **Google chat** channel

# Governance - **Weekly**

### Attendee

- Olenbee project manager
- Devoteam G cloud project manager
- Other stakeholders if required for technical support

### Objectives

- Operational management of the project
- Delivery tracking and planning updates
- Deliverables follow-up

### Deliverables

- Presentation slides
- Planning overview
- Detailed schedule
- Meeting minutes

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General Kick-off

# Delivery

- Workshops
- Deliverables

# Delivery - **Workshop IAM & Governance**

## Objective

- Present Cloud IAM & Gouvernance products and architectures
- Authenticating user and service identities.
- Ensuring proper authorization for actions and resource access.
- Organizing and labeling of cloud resources to ensure a structured, consistent and controlled environment.

## Subtopics

- Organization setup
- Users and groups management
- LDAP/Active Directory syncing
- Authentication and authorization
- Integration of single sign-on (SSO)
- Organization of folders and projects

## Out-of-scope

- Set up and run GCDS and Google GSPS on the customer's side.
- Configure SSO settings on the customer's side.
- Multi-region configurations.
- Resolve conflicting accounts.
- Naming conventions for resources.

## Mandatory attendees

- Infrastructure Architects from the customer
- Cloud Architects from the customer
- CISO or a member of the customer's Security Team

## Requirements

Customer teams have completed training courses such as 'Architecting with Google Compute Engine' or equivalent, and have experience using Google Cloud Platform (GCP) and related tools

## Key Success Factors

- Team availability
- Customer must answer all questions asked during the workshops within 72 times.

## Deliverables

- Updated version of the HLD document
- Updated version of the LLD document

# Delivery - **Workshop Networking**

## Objective

- Present Google Cloud Networking products and architectures
- Network architecture and main decision points
- Network security

## Subtopics

- Hub-and-spoke network design
- DNS topology
- Internet egress and traffic filtering

## Out-of-scope

- Multi-region configurations.
- Setup and integration of third-party solutions like HTTP proxies, firewalls, VPN appliances, DNS servers ...
- Naming conventions for resources.
- Hybrid connectivity options

## Mandatory attendees

- Infrastructure Architects from the customer
- Cloud Architects from the customer
- Network administrators from the customer
- CISO or a member of the customer's Security Team

## Requirements

Customer teams have completed training courses such as 'Architecting with Google Compute Engine' or equivalent, and have experience using Google Cloud Platform (GCP) and related tools

## Key Success Factors

- Team availability
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## Deliverables

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# Delivery - **Workshop Security**

## Objective

- Familiarize with security products offered by Google Cloud.
- Understand the shared responsibility model for security between cloud providers and client organizations.
- Define traceability measures.
- Strengthen resource protection.
- Improve data protection.

## Subtopics

- Data encryption and key management control levels.
- Security Command Center setup.
- Data access and lifecycle control.
- Organization policies.

## Out-of-scope

- Setup and integration of third-party solutions.
- Custom configuration of the Security Command Center.
- Setup and integration of the Chronicle and Secops platforms.
- Custom threat detection and remediation.
- Custom organization policies.
- Naming conventions for resources.

## Mandatory attendees

- Infrastructure Architects from the customer
- Cloud Architects from the customer
- Network administrators from the customer
- CISO or a member of the customer's Security Team

## Requirements

Customer teams have completed training courses such as 'Architecting with Google Compute Engine' or equivalent, and have experience using Google Cloud Platform (GCP) and related tools

## Key Success Factors

- Team availability
- Customer must answer all questions asked during the workshops within 72 times.

## Deliverables

- Updated version of the HLD document
- Updated version of the LLD document

# Delivery - **Workshop Billing**

## Objective

- Familiarize with billings products offered by Google Cloud.
- Help understanding billing data for cost controls

## Subtopics

- Billing data export
- Configure Data Studio for Billing Visualization
- Configure mandatory labels
- Configure simple Billing alerts

## Out-of-scope

- Export billing data to third-party solutions.
- Build custom dashboard or custom alerts.
- Naming conventions for resources.

## Mandatory attendees

- Infrastructure Architects from the customer
- Cloud Architects from the customer

## Requirements

Customer teams have completed training courses such as 'Architecting with Google Compute Engine' or equivalent, and have experience using Google Cloud Platform (GCP) and related tools

## Key Success Factors

- Team availability
- Customer must answer all questions asked during the workshops within 72 times.

## Deliverables

- Updated version of the HLD document
- Updated version of the LLD document

# Delivery - **Workshop Observability**

## Objective

- Familiarize with logging and monitoring products offered by Google Cloud.
- Centralized logging and monitoring

## Subtopics

- Determine scope, target container and retention for logs
- Determine access scopes to logging and monitoring information
- Optimizing logging scope with filters

## Out-of-scope

- Export logs to third-party logging solutions such as Elastic, Datadog...
- Export logs or metrics to third-party SIEM solutions.
- Export metrics to third-party monitoring solutions such as Zabbix, Dynatrace, Instana...
- Build custom alerts.
- Create dashboards, custom metrics, and queries.
- Naming conventions for resources.

## Mandatory attendees

- Infrastructure Architects from the customer
- Cloud Architects from the customer
- Network administrators from the customer
- CISO or a member of the customer's Security Team

## Requirements

Customer teams have completed training courses such as 'Architecting with Google Compute Engine' or equivalent, and have experience using Google Cloud Platform (GCP) and related tools

## Key Success Factors

- Team availability
- Customer must answer all questions asked during the workshops within 72 times.

## Deliverables

- Updated version of the HLD document
- Updated version of the LLD document

# Delivery - **Workshop Automation**

## Objective

- Understand the current DevOps platform used by the client.
- Present Infrastructure as Code tools selected for the project.
- Implement version control.
- Build CD pipelines for deploying the foundation.

## Subtopics

- Determine the Git repository for version control.
- Select pipeline tools to use in the following list: Jenkins, Gitlab Ci, Cloud Build and Github
- Build continuous deployment (CD) pipelines.
- Deploy the foundation.

## Out-of-scope

- Non-git repositories
- Custom pipelines and pipeline tools
- Installation of new pipelines tools
- Installing and configuring Running/Agents for the pipeline tools.
- Naming conventions for resources.

## Mandatory attendees

- Infrastructure Architects from the customer
- Cloud Architects from the customer
- Cloud Engineers from the customer

## Requirements

Customer teams have completed training courses such as 'Architecting with Google Compute Engine' or equivalent, and have experience using Google Cloud Platform (GCP) and related tools

## Key Success Factors

- Team availability
- Customer must answer all questions asked during the workshops within 72 times.

## Deliverables

- Final version of the HLD document
- Final version of the LLD document
- CD Pipelines
- GCP foundation

# Delivery - **Deliverables**

**Kickoff**
**& weekly decks**

Initialization & Weekly project management:

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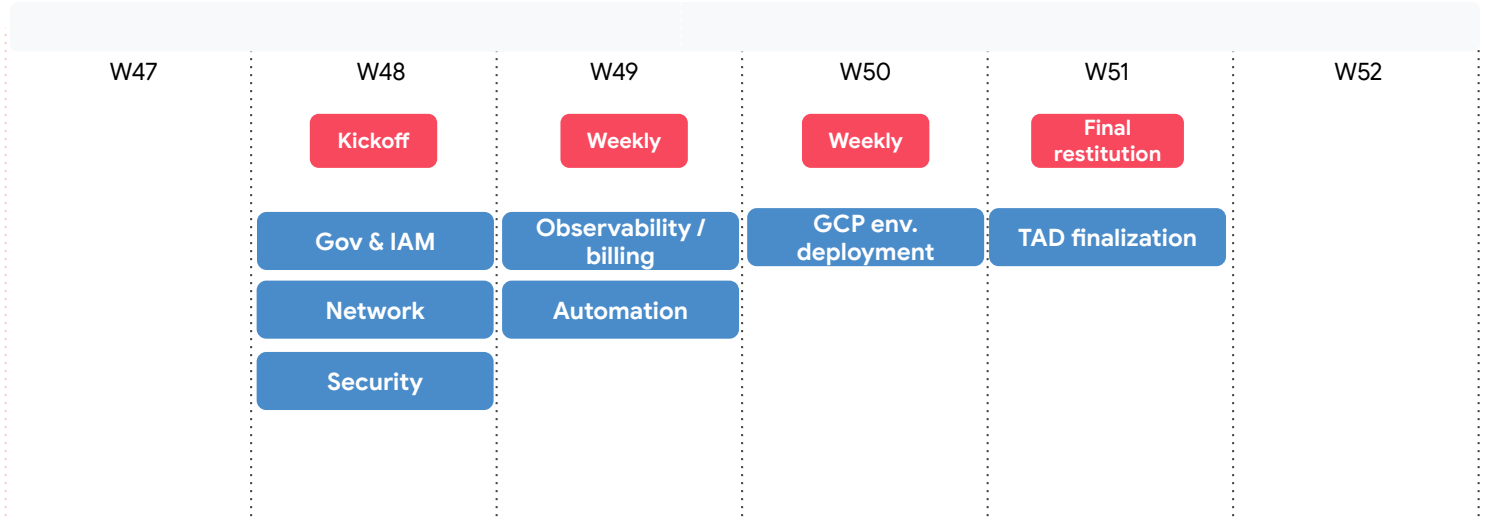
General Kick-off

# Planning

- Timeline
- Meetings

# Project organization.



# Planning - **Meetings**

## Weekly

### Project committee weekly *(30 min)*
Date and time: **12/04 & 12/11 15h-15h30 (Paris time)**
Attendees: **François, Houssem, Aymeric**

## Final restitution

### Final restitution *(1 h)*
Date and time: **12/18 15h-16h (Paris time)**
Attendees: **Martin, Francisco, Daniel, François, Houssem, Hae Jin, Aymeric**

## Workshops (2h)

### Workshop IAM & Gouvernance :
Date and time: **11/26 15h-17h (Paris time)**
Attendees: **Francisco, Daniel, François, Houssem, Hae Jin**

### Workshop Networking :
Date and time: **11/27 15h-17h (Paris time)**
Attendees: **Francisco, Daniel, François, Houssem, Hae Jin**

### Workshop Security :
Date and time: **11/29 15h-17h (Paris time)**
Attendees: **Francisco, Daniel, François, Houssem, Hae Jin**

### Workshop observability, billing :
Date and time: **12/03 15h-17h (Paris time)**
Attendees: **Martin, Francisco, Daniel, François, Houssem, Hae Jin**

### Workshop Automation :
Date and time: **12/05 15h-17h (Paris time)**
Attendees: **Martin, Francisco, Daniel, François, Houssem, Hae Jin**

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General Kick-off

# Next steps

- Prerequisites

# Next steps - **Prerequisites**

## GCP access

- An existing GCP Organization
- Creation of Cloud Identity users for devoteam engineers
- Grant organization admin role at organization level
- Grant billing account user role at billing account level
- Grant access to Git repository

## Documentation

- Share existing technical documentation

## Stakeholders

- Identify relevant stakeholders for each workshop